

Collapsible Wood Stool



Owen Palmer

Design for Disassembly

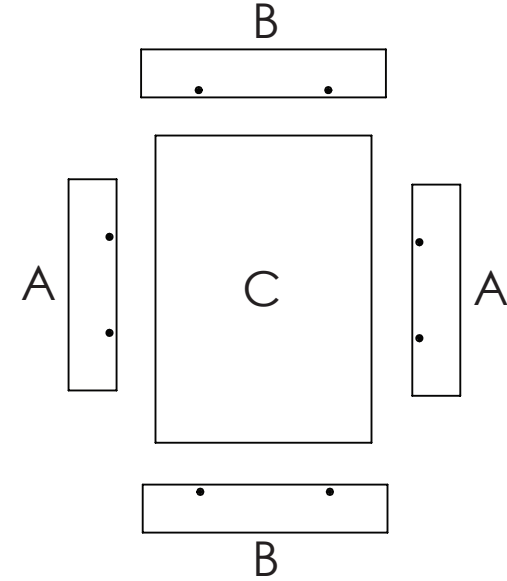
02/24/2025

Pre drill all marked holes to 1/8"
3" from the side 3/8" from the top
To avoid splitting the wood

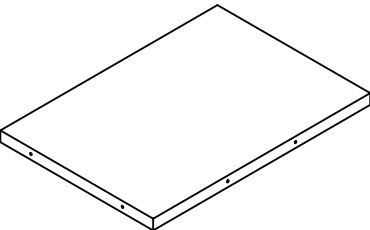
Top



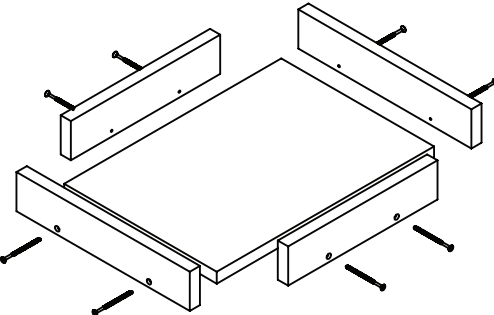
- A 1 X 3 X 11"
- B 1 X 3 X 12 3/4"
- C 1 X 12 X 16"
- #8 3" Screw X8



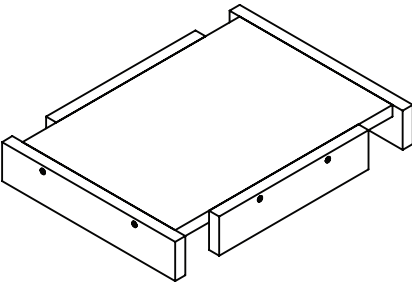
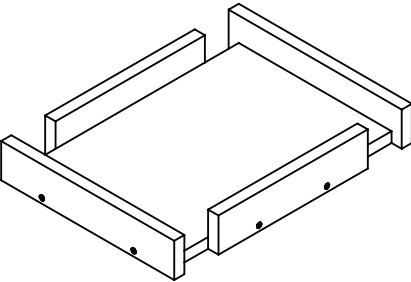
1



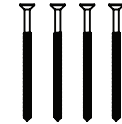
2



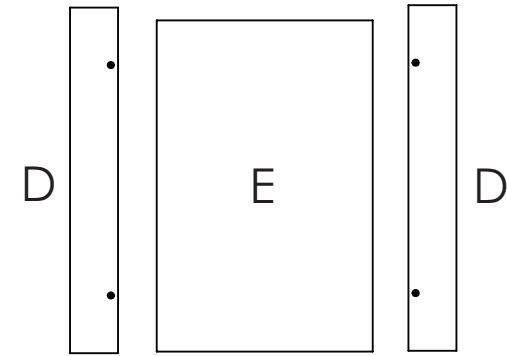
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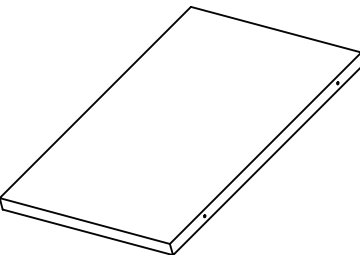
Side



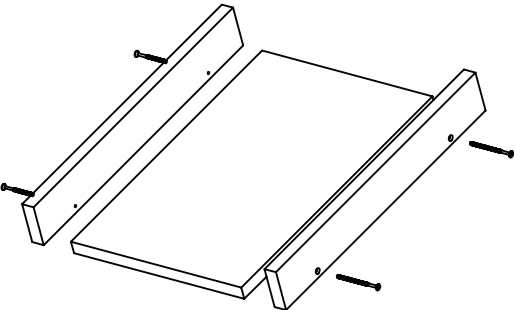
- D 1 X 3 X 18"
- E 1 X 12 X 17 1/4"
- #8 3" Screw X4



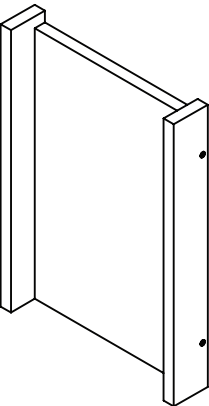
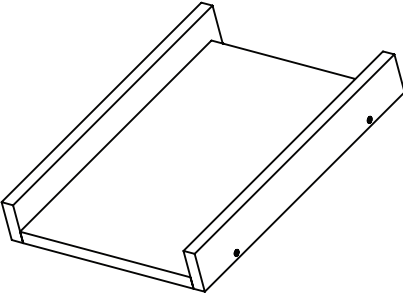
1



2



3

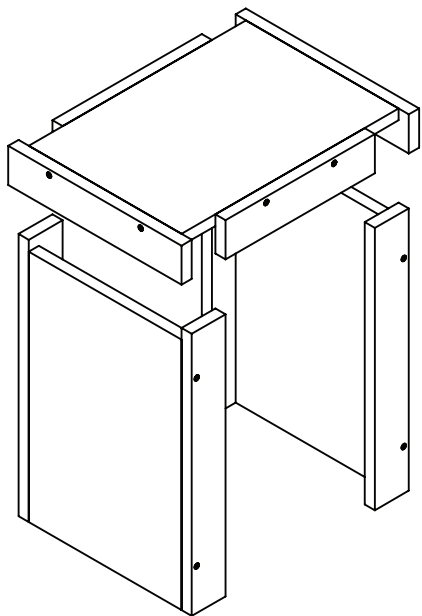


X2

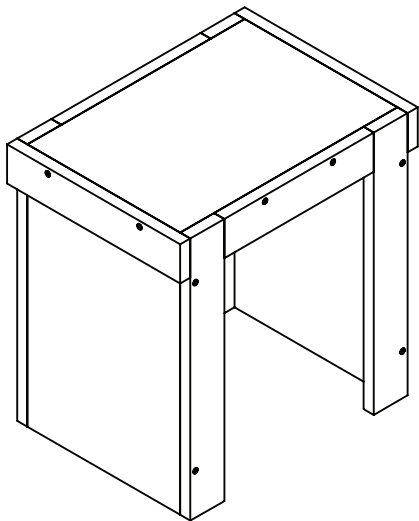
Final Assembly

Top x1

Side x2



Sand 12" edge of part C to ensure a tight fit between part A, B, and sides (use screws to adjust fit)



Finish as desired

Sand from 80 grit through 320 grit to ensure a smooth, even finish

Stain or sealant can be added after sanding to protect or achieve a desired finish

Bill of Materials

Part	Description	Dimensional Lumber	Length	Quantity
A	T Front/Back	1 X 3	11"	2
B	T Side	1 X 3	12 ¾"	2
C	T Centre	1 X 12	16"	1
D	S Side	1 X 3	18"	4
E	S Centre	1 X 12	17 ¼"	2
#8 3" Screw	Wood screw	-	3"	16

This stool is designed for disassembly in more ways than one. By only using screws as fasteners all parts of the stool can be disassembled easily back to their original cut sizes. The stool can also be easily disassembled into 3 components for storage or transportation, further adding value to the material and design. I wanted to limit the different types of dimensional lumber needed and limit cuts while still making an interesting design which could be collapsed. I was able to limit my design to only 2 types of lumber and 5 different pieces, only using straight cuts, while still maintaining functionality and stability. The pieces are very easy to line up by using other parts and a flat surface as a reference. The only thing I would really like to improve is the amount of waste material. Some of it could be used for a removable shelf, which I wanted to explore but did not have time for. There would still be a few feet of leftover material. I am very happy with the final product, it is incredibly sturdy, reasonably lightweight, looks good and can be collapsed.

